

```
In [8]: import pandas as pd
import numpy as np
df=pd.read_csv(r"C:\Users\Janvi Kapse\Downloads\patients - patients.csv")
print(df)
```

	PatientID	Name	Age	Gender	Weight	Disease
0	101	Amit	45.0	Male	72.0	Diabetes
1	102	Priya	NaN	Female	65.0	Hypertension
2	103	Rahul	120.0	Male	80.0	Asthma
3	104	Neha	35.0	Female	NaN	Diabetes
4	105	Vikas	52.0	Male	78.0	Hypertension
5	106	Sneha	28.0	Female	55.0	Fever
6	107	Rohan	41.0	Male	82.0	Diabetes
7	108	Pooja	33.0	Female	60.0	Asthma
8	109	Ankit	47.0	Male	76.0	Hypertension
9	110	Kavya	29.0	Female	58.0	Fever
10	111	Amit	45.0	Male	72.0	Diabetes
11	112	Meera	38.0	Female	62.0	Asthma
12	113	Karan	150.0	Male	90.0	Hypertension
13	114	Anjali	26.0	Female	54.0	Fever
14	115	Raj	49.0	Male	NaN	Diabetes
15	116	Simran	NaN	Female	59.0	Asthma
16	117	Deepak	44.0	Male	77.0	Hypertension
17	118	Ritu	31.0	Female	56.0	Fever
18	119	Arjun	53.0	Male	81.0	Diabetes
19	120	Nisha	36.0	Female	63.0	Asthma
20	121	Janvi	18.0	Female	52.0	Fever
21	122	Bhavika	26.0	Female	45.0	Diabetes
22	123	Chanchal	24.0	Female	60.0	Asthma
23	124	Kalash	24.0	Female	49.0	Hypertension
24	125	Hiteshri	21.0	Female	35.0	Diabetes
25	126	Mayur	20.0	Male	58.0	Fever
26	127	Aayush	29.0	Male	64.0	Asthma
27	128	om	24.0	Male	NaN	Diabetes
28	129	Anita	26.0	Female	57.0	Fever
29	130	Vilas	34.0	Male	NaN	Fever

```
In [9]: df.head()
```

```
Out[9]:
```

	PatientID	Name	Age	Gender	Weight	Disease
0	101	Amit	45.0	Male	72.0	Diabetes
1	102	Priya	NaN	Female	65.0	Hypertension
2	103	Rahul	120.0	Male	80.0	Asthma
3	104	Neha	35.0	Female	NaN	Diabetes
4	105	Vikas	52.0	Male	78.0	Hypertension

```
In [10]: df.tail()
```

```
Out[10]:
```

	PatientID	Name	Age	Gender	Weight	Disease
25	126	Mayur	20.0	Male	58.0	Fever
26	127	Aayush	29.0	Male	64.0	Asthma
27	128	om	24.0	Male	NaN	Diabetes
28	129	Anita	26.0	Female	57.0	Fever
29	130	Vilas	34.0	Male	NaN	Fever

```
In [4]: df.shape
```

```
Out[4]: (30, 6)
```

```
In [11]: df.columns.tolist()
```

```
Out[11]: ['PatientID', 'Name', 'Age', 'Gender', 'Weight', 'Disease']
```

```
In [12]: df.isnull().sum()
```

```
Out[12]: PatientID    0
         Name        0
         Age         2
         Gender      0
         Weight      4
         Disease     0
         dtype: int64
```

```
In [13]: df.fillna(df.mean(numeric_only=True),inplace=True)
         print(df.isnull().sum())
```

```
PatientID    0
Name         0
Age          0
Gender       0
Weight       0
Disease      0
dtype: int64
```

```
In [14]: print("Duplicated Records:",df.duplicated().sum())
```

```
Duplicated Records: 0
```

```
In [20]: Q1 = df["Age"].quantile(0.25)
         Q3 = df["Age"].quantile(0.75)
         IQR = Q3 - Q1
         lower_limit = Q1 - 1.5 * IQR
         upper_limit = Q3 + 1.5 * IQR
         print("Q1:",Q1)
         print("Q3:",Q3)
         print("IQR:",IQR)
         print("Lower limit:",lower_limit)
         print("Upper limit:",upper_limit)
```

Q1: 26.0
Q3: 44.75
IQR: 18.75
Lower limit: -2.125
Upper limit: 72.875

```
In [21]: outliers = df[(df["Age"] < lower_limit) | (df["Age"] > upper_limit)]  
print(outliers)
```

	PatientID	Name	Age	Gender	Weight	Disease
2	103	Rahul	120.0	Male	80.0	Asthma
12	113	Karan	150.0	Male	90.0	Hypertension

```
In [23]: df_clean = df[(df["Age"] <= lower_limit) & (df["Age"] >= upper_limit)]  
print("Shape Before:", df.shape)  
print("Shape After:", df_clean.shape)
```

Shape Before: (30, 6)
Shape After: (0, 6)

```
In [26]: df_clean.to_csv("Patient_cleaned.csv", index=False)  
print("Cleaned dataset saved successfully.")
```

Cleaned dataset saved successfully.

```
In [27]: print(df)
```

	PatientID	Name	Age	Gender	Weight	Disease
0	101	Amit	45.0	Male	72.000000	Diabetes
1	102	Priya	41.0	Female	65.000000	Hypertension
2	103	Rahul	120.0	Male	80.000000	Asthma
3	104	Neha	35.0	Female	63.846154	Diabetes
4	105	Vikas	52.0	Male	78.000000	Hypertension
5	106	Sneha	28.0	Female	55.000000	Fever
6	107	Rohan	41.0	Male	82.000000	Diabetes
7	108	Pooja	33.0	Female	60.000000	Asthma
8	109	Ankit	47.0	Male	76.000000	Hypertension
9	110	Kavya	29.0	Female	58.000000	Fever
10	111	Amit	45.0	Male	72.000000	Diabetes
11	112	Meera	38.0	Female	62.000000	Asthma
12	113	Karan	150.0	Male	90.000000	Hypertension
13	114	Anjali	26.0	Female	54.000000	Fever
14	115	Raj	49.0	Male	63.846154	Diabetes
15	116	Simran	41.0	Female	59.000000	Asthma
16	117	Deepak	44.0	Male	77.000000	Hypertension
17	118	Ritu	31.0	Female	56.000000	Fever
18	119	Arjun	53.0	Male	81.000000	Diabetes
19	120	Nisha	36.0	Female	63.000000	Asthma
20	121	Janvi	18.0	Female	52.000000	Fever
21	122	Bhavika	26.0	Female	45.000000	Diabetes
22	123	Chanchal	24.0	Female	60.000000	Asthma
23	124	Kalash	24.0	Female	49.000000	Hypertension
24	125	Hiteshri	21.0	Female	35.000000	Diabetes
25	126	Mayur	20.0	Male	58.000000	Fever
26	127	Aayush	29.0	Male	64.000000	Asthma
27	128	om	24.0	Male	63.846154	Diabetes
28	129	Anita	26.0	Female	57.000000	Fever
29	130	Vilas	34.0	Male	63.846154	Fever

```
In [28]: Q1 = df["Weight"].quantile(0.25)
Q3 = df["Weight"].quantile(0.75)
IQR = Q3 - Q1
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR
print("Q1:",Q1)
print("Q3:",Q3)
print("IQR:",IQR)
print("Lower limit:",lower_limit)
print("Upper limit:",upper_limit)
```

```
Q1: 57.25
Q3: 72.0
IQR: 14.75
Lower limit: 35.125
Upper limit: 94.125
```

```
In [29]: outliers = df[(df["Weight"] < lower_limit) | (df["Weight"] > upper_limit)]
print(outliers)
```

	PatientID	Name	Age	Gender	Weight	Disease
24	125	Hiteshri	21.0	Female	35.0	Diabetes

```
In [30]: df_clean = df[(df["Age"] <= lower_limit) & (df["Age"] >= upper_limit)]
print("Shape Before:", df.shape)
```

```
print("Shape After:", df_clean.shape)
```

Shape Before: (30, 6)

Shape After: (0, 6)

```
In [31]: df_clean.to_csv("Patient_cleaned.csv", index=False)  
         print("Cleaned dataset saved successfully.")
```

Cleaned dataset saved successfully.

In []: